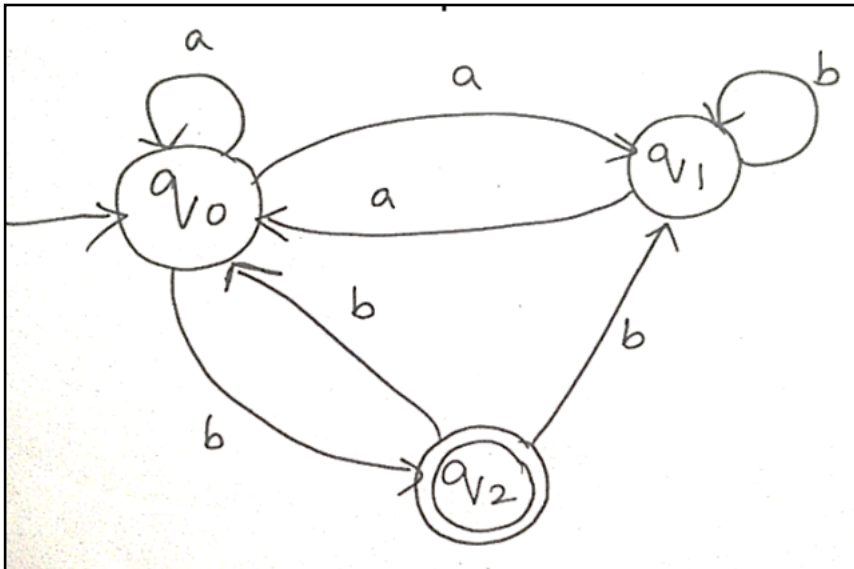


CO1_AT1 Analytical

1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L



Answer:

From the NFA diagram:

- States $Q = \{q_0, q_1, q_2\}$
- Alphabet $\Sigma = \{a, b\}$
- Start state = q_0
- Final state = q_2

Step 1: NFA Transition Table

State	A	B
q_0	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
$*q_2$	$\emptyset$	$\{q_0, q_1\}$

Step 2: Subset Construction

Start state of DFA = $\{q_0\}$

Find transitions for each subset.

State A = $\{q_0\}$

- on **a** = $\{q_0, q_1\}$
- on **b** = $\{q_2\}$

State B = $\{q_0, q_1\}$

- on **a**
 - $q_0 \rightarrow \{q_0, q_1\}$
 - $q_1 \rightarrow \{q_0\}$
 - Result = $\{q_0, q_1\}$
- on **b**
 - $q_0 \rightarrow \{q_2\}$
 - $q_1 \rightarrow \{q_1\}$
 - Result = $\{q_1, q_2\}$

State C = $\{q_2\}$

- on **a** = $\emptyset$
- on **b** = $\{q_0, q_1\}$

State D = $\{q_1, q_2\}$

- on **a**
 - $q_1 \rightarrow \{q_0\}$
 - $q_2 \rightarrow \emptyset$
 - Result = $\{q_0\}$
- on **b**
 - $q_1 \rightarrow \{q_1\}$
 - $q_2 \rightarrow \{q_0, q_1\}$
 - Result = $\{q_0, q_1\}$

Dead State = $\emptyset$

- on a $\rightarrow \emptyset$
- on b $\rightarrow \emptyset$

Step 3: DFA Transition Table

DFA State	a	b	Final?
$\rightarrow A = \{q_0\}$	B	C	No
$B = \{q_0, q_1\}$	B	D	No
$*C = \{q_2\}$	$\emptyset$	B	Yes
$*D = \{q_1, q_2\}$	A	B	Yes
$\emptyset$	$\emptyset$	$\emptyset$	No

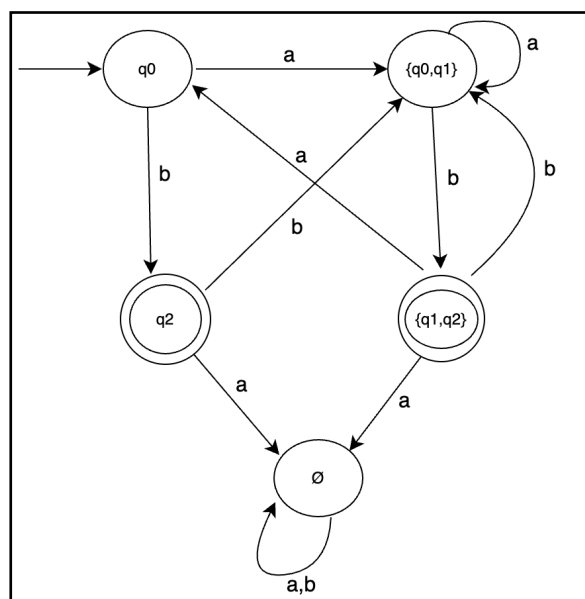
Final DFA states: Any subset containing q_2 .

Therefore,

- $\{q_2\}$
- $\{q_1, q_2\}$

are accepting states.

Step 4: DFA State Diagram



2. Let L be the language of words of length atleast 2 whose last two letters are difference from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA

Answer:

Step 1: NFA Construction

States:

- **q0** – Start state
- **q1** – Last symbol read is **a**
- **q2** – Last symbol read is **b**
- **q3** – Accept state (last two letters are **ab**)
- **q4** – Accept state (last two letters are **ba**)

NFA Transition Table

State	a	b
→q0	q1	q2
q1	q1	q3
q2	q4	q2
*q3	q1	q3
*q4	q4	q2

Step 2: Equivalent DFA

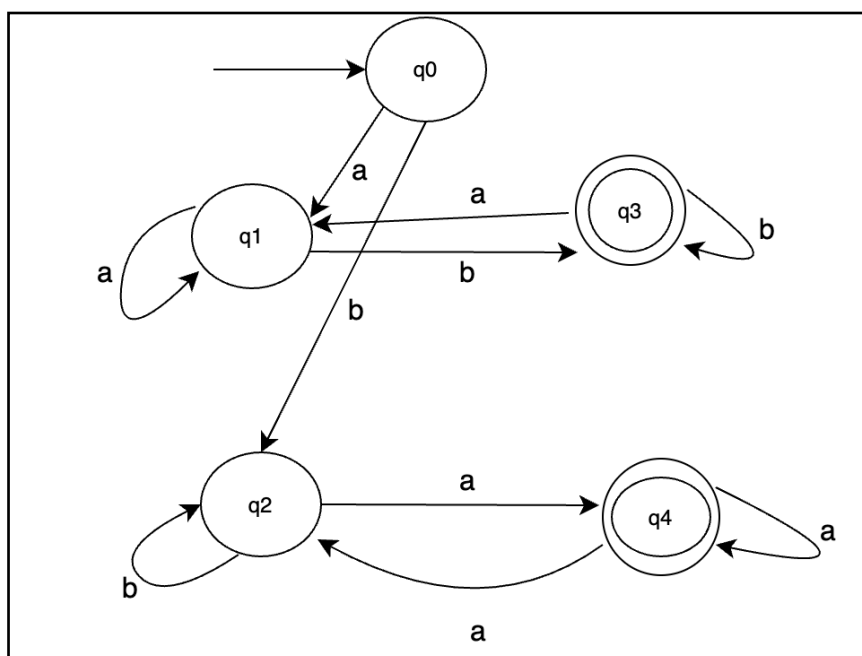
Since every state has exactly one transition for each input, this NFA is already deterministic.

Therefore,

Equivalent DFA = Same Automaton

State	a	b
$\rightarrow q_0$	q1	q2
q1	q1	q3
q2	q4	q2
*q3	q1	q3
*q4	q4	q2

DFA Diagram



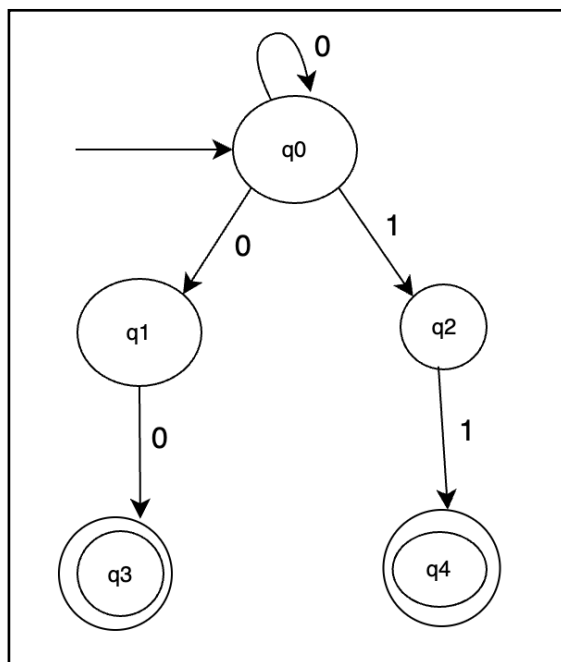
3.Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string 01011 is accepted by the NFA

Answer:

The NFA is defined as: $M=(Q,\Sigma,\delta,q_0 ,F)$
where

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0,1\}$
- q_0 = Start state
- $F = \{q_3, q_4\}$

NFA State Diagram



Transition Table

Current State	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\emptyset$	$\{q_4\}$
$*q_3$	$\emptyset$	$\emptyset$
$*q_4$	$\emptyset$	$\emptyset$

Acceptance of String 01011

Input: 01011

Input Read	State
Start	q ₀
0	q ₀
1	q ₀
0	q ₀
1	q ₂
1	q ₄ (Accept)

Since the NFA reaches the accepting state **q₄** after consuming the entire input, the string **01011** is **accepted**.

String: 01011

Computation:

$q_0 \rightarrow q_0 \rightarrow q_0 \rightarrow q_0 \rightarrow q_2 \rightarrow q_4$

Result: Accepted

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma=\{0,1\}$. Construct a DFA that represents the same language.

Step 1: NFA

States:

- **q0** – Start state
- **q1** – Found first 0
- **q2** – Found 00
- **q3** – Found 001 (Accept)

NFA Transition Table:

State	0	1
$\rightarrow q0$	{q0,q1}	{q0}
Q1	{q2}	$\emptyset$
q2	$\emptyset$	{q3}
*q3	{q3}	{q3}

Step 2: DFA using Subset Construction

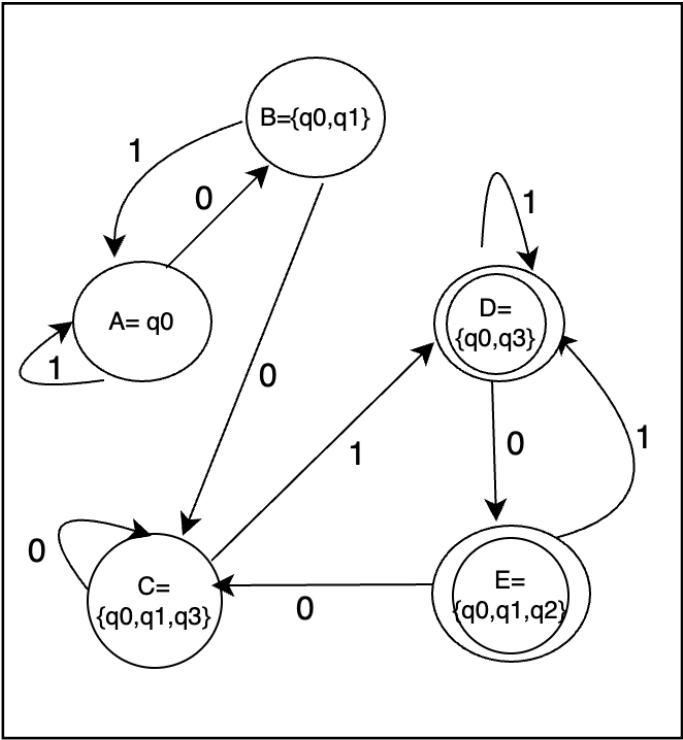
DFA States

- $A = \{q0\}$
- $B = \{q0,q1\}$
- $C = \{q0,q1,q2\}$
- $D = \{q0,q3\}$ (Accept)
- $E = \{q0,q1,q3\}$ (Accept)

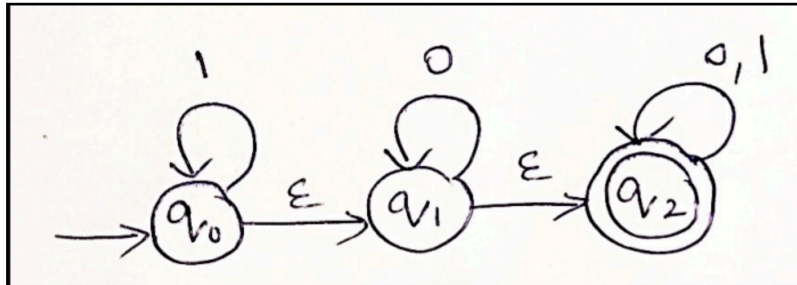
DFA Transition Table

DFA State	0	1
$\rightarrow A = \{q_0\}$	B	A
$B = \{q_0, q_1\}$	C	A
$C = \{q_0, q_1, q_2\}$	C	D
$*D = \{q_0, q_3\}$	E	D
$*E = \{q_0, q_1, q_3\}$	C	D

Step 3: DFA Diagram



5. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



From the given ϵ -NFA, we have:

States

- q_0 (Start)
- q_1
- q_2 (Final)

Transitions

- $q_0 \xrightarrow{1} q_0$
- $q_0 \xrightarrow{\epsilon} q_1$
- $q_1 \xrightarrow{0} q_1$
- $q_1 \xrightarrow{\epsilon} q_2$
- $q_2 \xrightarrow{0,1} q_2$

Step 1: ϵ -Closure of Each State

ϵ -Closure(q_0)

From q_0 :

- q_0
- $\epsilon \rightarrow q_1$
- $q_1 \xrightarrow{\epsilon} q_2$

Therefore,

$$\epsilon\text{-Closure}(q_0) = \{q_0, q_1, q_2\}$$

ϵ -Closure(q_1)

From q_1 :

- q_1
- $\varepsilon \rightarrow q_2$

Therefore,

$$\varepsilon\text{-Closure}(q_1) = \{q_1, q_2\}$$

$$\varepsilon\text{-Closure}(q_2)$$

No ε -transition.

Therefore,

$$\varepsilon\text{-Closure}(q_2) = \{q_2\}$$

Step 2: Construct Equivalent DFA

DFA Start State

$$\text{Start} = \varepsilon\text{-Closure}(q_0)$$

$$A = \{q_0, q_1, q_2\}$$

Since q_2 is present, **A is a final state.**

$$\text{From } A = \{q_0, q_1, q_2\}$$

Input 0

$$q_0 \rightarrow \emptyset$$

$$q_1 \rightarrow q_1$$

$$q_2 \rightarrow q_2$$

$$\text{Union} = \{q_1, q_2\}$$

$$\varepsilon\text{-closure} = \{q_1, q_2\}$$

$$B = \{q_1, q_2\}$$

Input 1

$$q_0 \rightarrow q_0$$

$$q_1 \rightarrow \emptyset$$

$$q_2 \rightarrow q_2$$

$$\text{Union} = \{q_0, q_2\}$$

$$\varepsilon\text{-closure}(\{q_0, q_2\})$$

$$q_0 \Rightarrow \{q_0, q_1, q_2\}$$

$$q_2 \Rightarrow \{q_2\}$$

Result:

$$A = \{q_0, q_1, q_2\}$$

$$\text{From } B = \{q_1, q_2\}$$

Input 0

$$q_1 \rightarrow q_1$$

$$q_2 \rightarrow q_2$$

$$\text{Result} = \{q_1, q_2\} = B$$

Input 1

$$q_1 \rightarrow \emptyset$$

$$q_2 \rightarrow q_2$$

$$\text{Result} = \{q_2\}$$

$$\varepsilon\text{-closure} = \{q_2\}$$

$$C = \{q_2\}$$

$$\text{From } C = \{q_2\}$$

$$\text{Input 0: } q_2 \rightarrow q_2 = C$$

$$\text{Input 1: } q_2 \rightarrow q_2 = C$$

DFA Transition Table

DFA State	0	1	Final
$\rightarrow^*A=\{q_0,q_1,q_2\}$	B	A	Yes
$*B=\{q_1,q_2\}$	B	C	Yes
$*C=\{q_2\}$	C	C	Yes

DFA Diagram

